Paul Mancion

Flexible CA/USA | Portfolio: paulmancion.com | linkedin.com/in/paulmancion
416-827-7996 | connect@paulmancion.com | github.com/MindMidas

Education

University of Guelph — Sep 2021 – May 2026
Honours. Bachelor of Computing, Computer Science (GPA 3.95/4.0) — *Guelph, ON*

University of Waterloo and Wilfrid Laurier — Sep 2020 – Apr 2021
Bachelor of Computer Science and Bachelor of Business Administration — *Waterloo, ON*

Technical Skills

Languages: Python, C, Java, SQL, JS, TypeScript, R | **Bilingual:** English, French
Libraries: PyTorch, TensorFlow, scikit-learn, Hugging Face, pandas, OpenCV, NumPy, Warp
Technologies: Docker, AWS, GCP, PostgreSQL, Git, FastAPI, OpenAI API, Cursor

Projects

Autopilot | AI-powered quant trading platform | *Python, scikit-learn, Hugging Face, CCXT, OpenAI, AWS*

- Built and validated an **end-to-end Machine Learning** trading pipeline that matched/outperformed the original strategy through signal prediction, hyperparameter **tuning**, and backtesting.
- Building an **LLM-driven trading automation platform** where **user-prompts** drive strategy creation, optimization, and deployment through **tool-calling**, **Model Context Protocol (MCP) servers**, and AST-validated code.
- Designing **async agent execution** with AWS Lambda, Step Functions, and isolated serverless workers for **scaling**.

CukeVision | Automated produce inspection | *Python, NumPy, pandas, PyTorch, OpenCV, Hugging Face*

- Selected ResNet18 model with **99.9% test accuracy** over a NN and Hugging Face MobileNetV2, and then built a **low-cost food-inspection app** with RMBG-2.0 + OpenCV object extraction, binary classification, and fine-tuning.
- Identified a **generalization gap** despite high test accuracy and improved classification by **curating** the Other class with **hard negatives** across shape, material, texture, scale, viewpoint, and lookalikes.

Chezz Engine | High-performance chess-variant AI | *C, Python, TypeScript, PostgreSQL*

- Increased move-generation speed by **~30x** and search depth for Chezz, a chess-variant game engine, by rewriting it from **Python to C** with bitboards, **bitwise operations**, precomputed move tables, and **magic attacks**.
- Built engine move selection with **multithreaded** alpha-beta Negamax, Zobrist hashing, transposition tables, and **custom heuristics** to **prune** non-competitive branches, avoid repeated evaluations, and search deeper.

NN Optimization | CIFAR-10 image classification | *Python, TensorFlow/Keras, scikit-learn, NumPy, pandas*

- Built a CIFAR-10 **image-classification pipeline** over 50,000 training and 10,000 test images, comparing Sigmoid FFNNs, ReLU DNNs, and **CNNs** using train/val/test error and **Macro F1**.
- Reduced **test error from 60.8% to 24.1%** by running a 48-configuration hyperparameter **grid search** across learning rate, hidden units, and dropout for a 3-block CNN.

Work Experience

Marcatus QED — Mar 2023 – May 2023
Marketing Communications Intern — *Toronto, ON*

- Created **Marcatus Mobile Education Platform** deck, showing impact across **77,000+** smallholder farmers.
- Developed the Marcatus **Unilever Global Summit 2023** deck, presenting expansion plans for **6,000+** farmers.

Docushield — May 2022 – Nov 2022
Product Growth Lead and Software Engineer Intern — *Los Angeles, CA*

- Drove **product-market fit** research, community initiatives, and IDO campaigns supporting a **$1M+** funding.
- Delivered **MVP** and **beta** releases for the core app and NFT platform by leading product design, **collaborating** with engineers across cross-functional teams, coordinating community **testing**, and supporting **CI/CD** release workflows.